CASE STUDY 51

BUS PASSENGER OCCUPANCY SIMULATION

Python Program:

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print("===== BUS PASSENGER OCCUPANCY SIMULATION =====")


capacity = int(input("Enter the maximum bus capacity: "))
passengers = 0
stops = int(input("Enter the number of stops: "))


if capacity <= 0 or stops <= 0:
    print("Error: Capacity and number of stops must be positive.")


else:
    for stop in range(1, stops + 1):
        print("\n--- Stop", stop, "---")


        entering = int(input("Passengers entering: "))
        leaving = int(input("Passengers leaving: "))


        if entering < 0 or leaving < 0:
            print("Invalid input: Passenger counts cannot be negative.")
            break


        if leaving > passengers:
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            print("Invalid input: More passengers leaving than onboard.")
            break

        passengers -= leaving

        if passengers + entering > capacity:
            print("Warning: Bus capacity would be exceeded.")
            available = capacity - passengers
            print("Maximum additional passengers allowed:", available)
            entering = available

        passengers += entering

        occupancy = (passengers / capacity) * 100

        print("Passengers onboard:", passengers)
        print("Occupancy: {:.2f}%".format(occupancy))

    else:
        print("\n===== FINAL BUS REPORT =====")
        print("Maximum capacity:", capacity)
        print("Final passengers onboard:", passengers)
        print("Available seats:", capacity - passengers)
        print("Final occupancy: {:.2f}%".format(
            (passengers / capacity) * 100
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))


print("\nSimulation completed.")
```

Output:

